

Introduction to Variables, Statistics and Machine Learning Features

Q1. Classification of 8 Daily-App Variables

Understanding the Concept

A **variable** is a characteristic, property, or piece of information that can have different values for different users, products, orders, or situations.

For example, in a food delivery application, **restaurant name, food price, delivery time, cuisine type, and payment method** are variables because their values can differ from one order or user to another.

Numerical Variables

A **numerical variable** represents a quantity or measurable amount. Its values are numbers, and mathematical calculations such as total, average, minimum, or maximum can generally be performed on them.

Examples:

- Number of Instagram likes = 250
- Product price = ₹999
- Delivery time = 35 minutes
- Number of followers = 1,500
- Product rating = 4.5

Categorical Variables

A **categorical variable** represents a group, category, class, or label. It tells us **what type or category something belongs to** rather than how much of something there is.

Examples:

- City = Ahmedabad
- Payment method = UPI
- Cuisine type = Gujarati
- Product category = Electronics
- Payment status = Paid

Difference Between Numerical and Categorical Variables

Numerical Variable	Categorical Variable
Represents quantity or measurement	Represents a group or category
Usually contains numerical values	Usually contains labels or names
Mathematical calculations are generally meaningful	Mathematical calculations are generally not meaningful
Example: ₹999	Example: UPI
Example: 35 minutes	Example: Gujarati cuisine

Classification of 8 Daily-App Variables

No	Variable	Example Value	Type	Reason
1	Number of likes	350 likes	Numerical	It represents a count that can be measured and compared.
2	Restaurant name	Domino's	Categorical	It identifies a restaurant rather than a quantity.
3	Product price	₹1,499	Numerical	Price represents a measurable monetary amount.

4	Payment method	UPI	Categorical	UPI, Cash, and Card are different categories.
5	Number of followers	2,500	Numerical	It represents a count of followers.
6	City name	Ahmedabad	Categorical	Cities are labels or categories.
7	Delivery time	30 minutes	Numerical	Time can be measured numerically in minutes.
8	Cuisine type	Gujarati	Categorical	Cuisine types represent different food categories.

Therefore, **number of likes**, **product price**, **number of followers**, and **delivery time** are numerical variables. **Restaurant name**, **payment method**, **city name**, and **cuisine type** are categorical variables.

Q2. Classification of Given Variables

Understanding the Variables

The question provides six variables:

1. Age of a user
2. City name
3. Payment status
4. Product rating
5. Delivery time in minutes
6. Cuisine type

To classify them, we determine whether each variable represents a **measurable quantity** or a **category/label**.

Classification

No	Variable	Example	Classification	Reasoning
1	Age of a user	20 years	Numerical	Age represents a measurable quantity and can be expressed numerically.
2	City name	Ahmedabad	Categorical	A city name is a label representing a geographical category.
3	Payment status	Paid	Categorical	Paid, Unpaid, and Pending represent different categories or states.
4	Product rating	4.5 stars	Numerical	A rating is represented by a numerical score that can be compared and averaged.
5	Delivery time	35 minutes	Numerical	Delivery time measures the amount of time taken for delivery.
6	Cuisine type	Gujarati	Categorical	Cuisine type identifies a category of food.

Detailed Reasoning

1. Age of a user — Numerical

Age measures how old a person is. For example, a user may be 18, 20, or 25 years old.

2. City name — Categorical

City name identifies a location category. Examples include Ahmedabad, Mumbai, and Delhi.

3. Payment status — Categorical

Payment status represents different states such as Paid, Unpaid, or Pending.

4. Product rating — Numerical

Product rating is represented by a numerical score such as 3.5, 4.0, or 4.5 stars. These values can be compared and an average rating can be calculated.

5. Delivery time — Numerical

Delivery time measures how long an order takes to arrive, such as 25, 40, or 60 minutes.

6. Cuisine type — Categorical

Cuisine type represents food categories such as Gujarati, Punjabi, Chinese, Italian, or South Indian.

Final Classification

Numerical Variables: Age, Product Rating, Delivery Time

Categorical Variables: City Name, Payment Status, Cuisine Type

Q3. Use of Statistics in a Cricket or IPL App

Understanding Statistics

Statistics is the process of collecting, organizing, analyzing, interpreting, and presenting data to understand patterns and make informed decisions.

A cricket or IPL application generates a large amount of data related to:

- Players
- Teams
- Matches
- Runs
- Wickets
- Scores
- Match results
- Player performance

Statistics can transform this raw data into useful information for users.

Feature 1: Player Runs

Player runs represent the total number of runs scored by a cricket player.

Example

Player	Runs
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Player A	650
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Player B	580
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Player C	475
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Player D	390
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How Statistics Can Be Used

Statistics can be used to calculate:

- Total runs
- Average runs per match
- Highest score
- Recent match performance
- Player comparisons
- Performance trends

For example, an application can compare the total runs scored by different players and display the highest run scorers.

How It Can Improve User Experience

Instead of making users check every individual scorecard, the application can provide:

- Top run scorers
- Recent-performance statistics
- Player comparison
- Average runs per match
- Highest scores
- Performance graphs

This helps cricket fans understand player performance quickly.

Feature 2: Team Name

Team name is a categorical variable that identifies the team playing a match or the team to which a player belongs.

Examples include:

- Gujarat Titans
- Mumbai Indians
- Chennai Super Kings
- Royal Challengers Bengaluru

Although the **team name itself is categorical**, statistics can be applied to the performance data associated with each team.

How Statistics Can Be Used

The application can calculate:

- Matches played
- Matches won
- Matches lost
- Win percentage
- Average team score
- Average wickets
- Recent performance

Example: Win Percentage

Suppose a team has played **10 matches** and won **7 matches**.

The formula is:

$$\text{Win Percentage} = \frac{\text{Matches Won}}{\text{Total Matches}} \times 100$$

Substituting the values:

$$= \frac{7}{10} \times 100 = 70\%$$

Therefore, the team's win percentage is **70%**.

How It Can Improve User Experience

The application can use team statistics to provide:

- Team performance summaries
- Points-table analysis

- Team comparisons
- Recent form
- Statistical match insights

Users can understand team performance quickly without manually calculating the statistics.

Conclusion

Statistics helps cricket applications transform raw player and team data into meaningful information. This can make the application more informative, interactive, and easier for users to understand.

Q4. Variables as Features in a Machine Learning Model

Understanding Machine Learning Features

In **Machine Learning (ML)**, a **feature** is an input variable or piece of information that a model uses to identify patterns and make predictions.

For example, a food delivery application may want to predict:

“Which type of food is this user likely to order next?”

The model could use information such as:

- Previous orders
- Preferred cuisine
- Average spending
- Order frequency
- Delivery-related information

These variables can be used as **features**.

Example

Feature	Example Value
Previous cuisine	Gujarati

Average order value	₹350
Orders per month	8
Previous rating	4.5
Delivery location	Ahmedabad

The machine learning model can analyze patterns in such data and generate predictions or recommendations.

Five Variables as ML Features

No	Variable	Example	ML Feature?	Why?
1	Previous order history	Pizza, Burger, Gujarati Thali	Yes	Past purchases can provide information about the user's interests and future ordering behavior.
2	Average order value	₹450	Yes	It can represent the user's typical spending level and may help personalize offers.
3	Preferred cuisine	Gujarati	Yes	It can help recommend restaurants or dishes matching the user's preferences.
4	Delivery time	35 minutes	Yes	Delivery-time patterns may provide information about the user's preference for faster delivery.

5	Payment method	UPI	Yes	Payment behavior can provide additional information for certain personalization or offer-related predictions.
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Detailed Explanation

1. Previous Order History

Previous order history can be a useful machine learning feature because past behavior may provide information about future behavior.

For example, if a user frequently orders pizza, the application may recommend similar pizza restaurants or dishes.

Possible use: Personalized food recommendations.

2. Average Order Value

Average order value represents approximately how much money a user normally spends per order.

For example:

- Order 1 = ₹300
- Order 2 = ₹400
- Order 3 = ₹500

Formula:

$$\text{Average Order Value} = \frac{300 + 400 + 500}{3} = \frac{1200}{3} = ₹400$$

Therefore, the average order value is **₹400**.

This information may help an application provide offers appropriate to the user's typical spending pattern.

Possible use: Personalized discounts and offers.

3. Preferred Cuisine

Preferred cuisine is a categorical variable.

Examples:

- Gujarati

- Punjabi
- Chinese
- Italian
- South Indian

A categorical variable can be converted into a suitable numerical representation using encoding techniques before being used by many machine learning models.

If a user frequently orders Gujarati food, the application may recommend Gujarati restaurants or dishes.

Possible use: Personalized food recommendations.

4. Delivery Time

Delivery time is a numerical variable and can be used as a feature when it is relevant to the prediction task.

For example, the application can analyze previous delivery-time patterns and identify whether delivery speed is associated with a user's choices.

Possible use: Delivery-related personalization.

5. Payment Method

Payment method is a categorical variable.

Examples include:

- UPI
- Credit Card
- Debit Card
- Cash on Delivery
- Wallet

It can be used as a feature when payment behavior is relevant to the prediction being made.

Possible use: Payment-related personalization or offers.

Important Concept: Feature vs Target

A **feature** is an input variable used by a machine learning model.

A **target** is the outcome that the model is trying to predict.

Concept	Meaning	Example
Feature	Input used by an ML model	Previous orders
Target	Output the model tries to predict	Next food category

Example

Suppose the application wants to predict:

“Will the user order pizza?”

Then:

Features:

- Previous order history
- Average order value
- Preferred cuisine

Target:

- Will order pizza? — Yes/No

Simple ML Structure

Features → Machine Learning Model → Prediction

Example:

Previous Orders + Cuisine Preference + Average Spending → ML Model → Recommended Food
